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Abstract

Sentiment analysis of Malaysian Malay-English text needs reliable labels for both training and testing. However, the four Malaysian resources discussed in this review provide limited evidence of label quality. This review discusses the language features that make the annotation difficult, compares methods of assigning labels, and examines how researchers determine agreement. One concern is whether a neutral label means that the message has no sentiment or it means that a tool has missed the sentiment. For example, 78.23% of MESocSentiment posts are labelled as neutral, and its published examples also include positive Malay-English expressions that automatic tools labelled as neutral. These examples show that further checking is needed. Besides that, another concern is that Cohen's kappa can be difficult to interpret when one label is much more common than the others. Future studies should explain their annotation procedures, retain individual labels, and report agreement for the task being studied.

Keywords: *sentiment analysis, code-mixed text, Manglish, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is a research field that aims at understanding the underlying sentiment of unstructured content (Poria et al., 2023). Organisations use it to understand public opinion in areas such as consumer products and services, healthcare, political elections and financial services. However, most language resources used to train language models are developed for individual languages only (Joshi et al., 2020). This becomes a problem when a sentence switches between two languages, which is known as code-switching. A tool may recognise an opinion in one language but miss it in another language. As a result, the performance on code-switched text is much worse than the performance on the same tasks in a monolingual setting (Doğruöz et al., 2021; Winata et al., 2023). This review uses sentiment tool for systems that assign labels directly, such as lexicon-based analysers, and language model for pretrained models that are fine-tuned for a task.

This problem is clearly visible in Malaysia. Malay-English code-switched text is common in Malaysian social media and written communication (AlAfnan et al., 2025). In this review, Manglish refers to the mixing of Malay and English. Manglish

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communication has recurring switching patterns (Treffers-Daller et al., 2022) and informal spellings, such as x for tidak, camne for macam mana (Maskat and Rahman, 2020). Because these forms are informal, standard language resources may not include them. Thus, a tool may miss an expression that a Malaysian reader can understand, including negative or positive words that can change the entire sentiment of the message (Abu Bakar et al., 2020).

Part of this difficulty comes from how multilingual language models are built and trained. Language models such as mBERT (Devlin et al., 2019) and XLM-RoBERTa (Conneau et al., 2020) are mainly trained on monolingual text and may have limited exposure to code-switched text (Doğruöz et al., 2021). Supporting several languages does not necessarily mean that a model handles those languages well when they appear together in a single message (Winata et al., 2021). Covering more languages can also place demands on the model's available capacity (Conneau et al., 2020). When one model shares limited capacity across many languages, each language gets less. Further training on code-switched examples can improve performance (Khanuja et al., 2020), but the results vary by language and training conditions (Adilazuarda et al., 2022).

The models are only one part of the problem. Data quality is not only about pretraining text. It also concerns the labels used to train and test these models. In supervised sentiment analysis, each training example has a label that tells the model which sentiment it should learn to recognise (Poria et al., 2023). Incorrect labels can teach the model the wrong patterns (Frénay and Verleysen, 2014). Not only that, labels also matter during testing. If the test labels are wrong, the reported results may be misleading and may even change which model appears to perform best (Northcutt et al., 2021). Reliable evaluation therefore depends on the quality of the labels, as well as the model itself (Bowman and Dahl, 2021). For Manglish, labelling is harder because annotators must judge informal spellings and mixed words.

These concerns motivate a closer look at the resources that are already available. This review examines four Malaysian Malay-English sentiment resources: MY-EN-CS, SentiBahasaRojak, MyCovid-Senti and MESocSentiment (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024). For general-purpose use, this review considers four desirable features: at least 10,000 labelled messages, a class distribution that is not dominated by one label, everyday social-media language, and direct evidence that the labels are reliable. These are the criteria used in this review, rather than universal requirements for every dataset. Each resource meets some of them, but none meets all four.

The class distribution is the first gap. MESocSentiment is the largest of these resources, but 78.23% of its posts are labelled neutral (Mohd Shamsuddin et al., 2024). The study also presents examples in which automatic tools miss positive Malay expressions. These examples need further investigation, although they do not show the overall error rate.

Evidence about label reliability is the second gap. The four resources provide limited information about agreement between annotators. MyCovid-Senti and MESocSentiment do not report an agreement coefficient (Kong et al., 2023; Mohd

Shamsuddin et al., 2024). MY-EN-CS and SentiBahasaRojak report agreement for tasks with two categories, rather than for the three sentiment labels of positive, negative and neutral (Kasmuri and Basiron, 2019; Romadhona et al., 2022).

Agreement is normally checked in a standard way. In order to check consistency, researchers can ask annotators to label the same messages independently. They can then calculate an agreement measure such as Cohen's kappa (κ), which accounts for agreement expected by chance (Cohen, 1960; Artstein and Poesio, 2008). However, Cohen's kappa value is affected by how often each annotator uses each label. This makes it harder to interpret when one class is much larger than the others (Feinstein and Cicchetti, 1990; Byrt et al., 1993; Di Eugenio and Glass, 2004). A dataset with many neutral labels therefore needs careful checks of both label quality and agreement. Besides these quality issues, Selvaraju et al. (2024) reported a shortage of annotated code-mixed Malay resources.

These gaps set the question for this review. It asks what prevents researchers from establishing reliable annotation and evaluation for Manglish sentiment data, focusing on language difficulties, labelling methods and evidence of label quality. The purpose here is to assess the evidence supporting existing labels and identify what future studies need to check. Reviews of sentiment classifiers and sentiment analysis more broadly are available elsewhere (Poria et al., 2023; Nazir et al., 2022). No new system is proposed.

Based on this question, the review makes three contributions:

- It compares manual annotation, existing sentiment tools and generative models, focusing on the effort required and the evidence of label reliability.
- It explains the difference between missed sentiment cues and reasonable differences in interpretation, and examines the reliability evidence for four Malay-English resources.
- It shows how language difficulties, labelling methods and class imbalance affect one another, and identifies priorities for future research.

The review is organised as follows. Section 2 describes the literature selection. Section 3 discusses language difficulties, followed by annotation approaches in Section 4 and reliability evaluation in Section 5. Section 6 brings these issues together and outlines research priorities. Section 7 concludes the review.

2. Review Methodology

The literature was identified through structured searches in the ACL Anthology, IEEE Xplore and Scopus. The searches combined terms from three groups. Language terms included code-mixed, code-switching, Manglish, Bahasa Rojak and Malay-English. Task terms included sentiment analysis, polarity and opinion mining. The final group focused on datasets and label quality, using corpus, annotation, inter-annotator agreement and label quality.

The main search covered studies published from 2019 to 2026. Earlier studies were included when they were cited in the papers found during the search. Studies on agreement measures were not limited to this period because several commonly used measures were introduced before 2019.

Studies had to meet four conditions to be included. They had to examine Malaysian Malay-English text or methods that could be applied to its annotation. They also had to explain how labels were produced and checked, appear in a peer-reviewed journal or conference, and include work on annotation or reliability. Preprints without a peer-reviewed version were excluded.

Studies that only reported classifier accuracy on an existing dataset were excluded. Such studies use the supplied labels to judge model performance, but do not necessarily check those labels. This review focuses on the evidence that the labels themselves are reliable.

3. Linguistic Challenges for Annotation

Three features of Malay-English text are especially relevant to annotation: switching between languages, particles whose meaning depends on context, and non-standard spelling.

3.1. Sentiment across language boundaries

Malay and English can appear within a single sentence. Treffers-Daller et al. (2022) found frequent switching within clauses in Malaysian classroom speech. However, the study does not establish how common this pattern is in social-media writing.

This matters for sentiment annotation because an opinion may be expressed in either language. A tool may understand the English part but miss an opinion expressed in Malay. Bilingual annotators can read both parts. However, they still need guidance when a message contains several targets of opinion, because such messages produce inconsistent labels without clear instructions (Mohammad, 2016).

3.2. Particles and context

Particles such as lah can affect how a message sounds without changing its basic content. Treffers-Daller et al. (2022) describe several uses, including expressing assertiveness and encouragement. AlAfnan et al. (2025) use "Don't disturb la, he's busy" as an example of a softened instruction. The spelling la here also shows the variation discussed in Section 3.3. These studies examine classroom speech and television dialogue, so they show what particles can do rather than how often each use appears on social media.

Lah therefore cannot be given a fixed positive or negative meaning. Depending on the context, an expression may sound irritated or polite. Annotation guidelines should explain how to consider this tone when deciding whether the message expresses a positive, negative or neutral opinion.

3.3. Spelling variations

Users often write Malay words in shortened or informal ways. Maskat and Rahman (2020) report forms such as jgn for jangan and mcmana for macam mana. These forms do not match the standard spellings stored in lexicons and pretrained

models, so the word may not be recognised at all. One response is spelling normalisation, which maps an informal form back to its standard spelling. In Maskat and Rahman's (2020) experiment, the highest precision was 0.43. This means that fewer than half of the proposed matches were correct. However, the evaluation excluded English words and code-mixed posts. The result therefore does not measure performance on Malay-English code-mixed text.

Negation makes spelling variation particularly relevant to sentiment. As mentioned in Section 1, x can mean tidak. If a tool or model fails to recognise this negative form, it may misunderstand the opinion. A spelling-normalisation method should therefore be checked for whether it keeps the original sentiment, as well as whether it produces a standard spelling.

4. Annotation Approaches and Their Limitations

4.1. Human and automatic labelling

A human annotator understands how the language is actually used locally. A human annotator can read context, informal spellings and particles, and judge the message as a whole. But manual work takes time and requires suitable language skills. Kasmuri and Basiron (2019) developed a subjectivity corpus from blogs. Kong et al. (2023) used three independent annotators and written guidelines to label 11,568 tweets. Human judgement does not always guarantee good labels, so the results still need to be checked for consistency.

Because manual work is slow, some datasets reuse existing labelled data instead of collecting new text. In SentiBahasaRojak, the product and movie subsets were created by automatically translating parts of existing Malay sentences into other languages, so the code-mixed sentences were machine-produced rather than written by users (Romadhona et al., 2022). The stock-market posts were crawled and annotated by five experts as positive or negative. The reported agreement covers only the stock-market subset. It should therefore be interpreted as evidence for that subset, rather than for the entire dataset.

Automatic tools can assign labels more quickly. Mohd Shamsuddin et al. (2024) used TextBlob, VADER and pysentimiento to label MESocSentiment. The label supported by the majority of the tools became the final label. The authors describe the overall dataset construction as semi-automatic, although the sentiment labels themselves were assigned automatically. Their accuracy on Malay-English text still needs to be checked separately.

Generative models offer another way to label text, but their performance needs to be tested for the language and task concerned. Jadhav et al. (2025) found that GPT-4o and Llama 3.1 405B performed worse than a fine-tuned BERT model on four Marathi classification tasks, including sentiment analysis. This finding does not establish how the models would perform on Malaysian text, and none of the studies reviewed here tests generative models as annotators for Malay-English sentiment.

4.2. Interpreting neutral labels

MESocSentiment contains 19,714 posts, of which 78.23% are labelled neutral, 16.33% positive and 5.44% negative (Mohd Shamsuddin et al., 2024). A large neutral class does not automatically mean that the labels are wrong. It may reflect the topics collected, the limited coverage of the labelling tools, or messages whose sentiment is genuinely uncertain. The question is whether some posts were labelled as neutral because the tools failed to recognise an opinion.

A neutral share of 78% is high compared with other Malaysian data. In Kong et al. (2023), three annotators labelled a COVID-19 Twitter dataset and 23.6% of the posts are neutral. The two datasets cover different content, so the gap is not a like-for-like measure. It does suggest, however, that the neutral share in MESocSentiment is worth investigating.

One possible cause of an incorrect neutral label is limited language coverage. VADER, one of the three tools used, calculates sentiment from the cues it recognises and assigns neutral when the resulting score is near zero (Hutto and Gilbert, 2014). If its resources do not include a Malay expression, that expression may not contribute to the score. A message with an opinion may then receive a neutral label.

The examples published with MESocSentiment support this explanation. Of the six labelled examples presented by Mohd Shamsuddin et al. (2024), three contain positive Malay expressions that were labelled neutral. One of the expressions is memang cantik sangat, meaning really very beautiful. These examples show that the tools can miss Malay sentiment. However, six examples are not enough to estimate the error rate for the whole dataset.

Not every neutral label reflects a missed cue. Plank (2022) argues that variation between annotators is not always noise, since a message may have more than one plausible reading. In a study of natural language inference, disagreement between raters persisted even when 50 judgements were collected for each item, and providing more context did not reduce it (Pavlick and Kwiatkowski, 2019). These studies do not measure Malaysian data, but they point to an annotation issue that MESocSentiment does not address. A message with no sentiment should be distinguished from one whose sentiment is uncertain.

4.3. Limits of majority voting

Majority voting weights all answers equally, so the final label is wrong whenever the majority is wrong (Hovy et al., 2013). In MESocSentiment, the voters are tools rather than people, but the same limit applies. Each of the three tools labels most retained posts as neutral, as shown in Table 1.

Table 1. Neutral labels in the retained MESocSentiment corpus

Tool	Neutral posts	Share
TextBlob	14,305	72.56%
VADER	13,115	66.53%
pysentimiento	16,916	85.81%

Tool	Neutral posts	Share
Majority vote	15,422	78.23%

Source: Mohd Shamsuddin et al. (2024). Percentages are calculated from the 19,714 retained posts.

The shares differ by tool. Pysentimiento has the highest neutral share. It is trained for English, Spanish, Portuguese and Italian, and was selected because it is trained on English tweets (Pérez et al., 2023; Mohd Shamsuddin et al., 2024). Malay is not among these languages. This share alone does not show whether the tool is less accurate. The totals also do not reveal whether the tools make mistakes on the same posts. If they do, the majority label would carry the same error. Bilingual reviewers would need to check the posts independently to answer that question.

The authors removed 512 of the original 20,226 posts because all three tools gave different labels (Mohd Shamsuddin et al., 2024). Each removed post therefore had one neutral vote, one positive vote and one negative vote. The number of neutral posts stayed at 15,422 while the total became smaller, so the neutral share rose from 76.24% to 78.23%. Removing disputed posts may also exclude difficult examples, although the published counts do not show what these posts contain. Keeping the posts and each tool's labels would allow researchers to examine the disagreements directly (Uma et al., 2021).

5. Evaluating Annotation Reliability

Agreement shows how consistently annotators assign the three sentiment labels, but it does not prove that the labels are correct. Two annotators may agree simply because they made the same mistake. Cohen's kappa (Cohen, 1960) addresses part of this problem. It measures how far the observed agreement exceeds the agreement expected by chance, where chance is estimated from how often each annotator uses each label. Artstein and Poesio (2008) discuss commonly used guidance that treats values above 0.8 as reliable, and values between 0.67 and 0.8 as supporting only tentative conclusions. These ranges should be read as guidance rather than fixed rules, because the task, the annotators, and the label distribution all affect what a given value means.

The label distribution deserves particular attention. When one label is far more common than the others, the agreement expected by chance rises, and kappa falls even if the annotators match on the same proportion of items. Di Eugenio and Glass (2004) illustrate this with two binary examples that both reach 90% observed agreement. Kappa is 0.80 for the balanced example, but −0.048 for the skewed one. Sentiment corpora are often skewed in this way, so observed agreement and kappa should be reported and read together.

This applies directly to sentiment data, where neutral often dominates. Overall agreement should therefore be reported alongside per-class agreement, so that weak agreement on the smaller classes is not hidden by the large one. The annotations must also be stored per annotator, not only as label totals, because Cohen's kappa needs to know which annotator chose which label (Di Eugenio and Glass, 2004). No single coefficient gives a complete account of reliability.

Table 2. Reliability evidence reported for the reviewed resources

Resource	Annotation evidence	Scope and limitation
MY-EN-CS Kasmuri and Basiron (2019)	Two bilingual undergraduates; written guideline; reported kappa = 0.83.	Opinionated versus factual judgement. Does not establish three-class polarity agreement.
SentiBahasaRojak Romadhona et al. (2022)	Five experts labelled stock posts; reported kappa = 0.77.	Positive versus negative stock labels; coefficient variant unspecified. Other subsets were generated automatically.
MyCovid-Senti Kong et al. (2023)	Three independent annotators; written guideline.	Three-class sentiment labels; no agreement coefficient reported.
MESocSentiment Mohd Shamsuddin et al. (2024)	Majority vote among three automatic tools.	No human inter-annotator agreement coefficient reported.

Table 2 summarises what each resource reports and what remains untested. MyCovid-Senti and MESocSentiment report no agreement coefficient at all. MY-EN-CS and SentiBahasaRojak do report kappa values above the common thresholds, but both figures come from two-category tasks, so neither shows that annotators agree on a three-way positive, negative and neutral distinction. SentiBahasaRojak also used five annotators; Cohen's kappa in its original form is defined for two coders, and generalised versions exist for larger groups, so a study of this kind should state which variant it applied. None of this shows that the four resources are unreliable. It shows that their reliability for three-class sentiment has not yet been demonstrated.

6. Research Priorities

The reviewed studies do not always give the same attention to checking labels as they give to producing them. The following discussion explains how the problems are connected and sets out four priorities for improving annotation and evaluation.

6.1. How the challenges interact

The challenges described above are not independent. Manglish is written with inconsistent spelling, and its discourse particles (short words such as lah and meh, which signal the speaker's attitude) depend heavily on context. Tools trained on other languages therefore miss many of these cues. Bilingual annotators can read

them, but hiring them is expensive. Mohd Shamsuddin et al. (2024) cite this cost when choosing automatic labelling instead. However, automation inherits the same blind spots: for example, VADER relies on an English sentiment lexicon (Hutto and Gilbert, 2014). When several tools miss the same expression, majority voting does not correct the error; it confirms it. Opinionated posts are then recorded as neutral, which both loses data and skews the label distribution discussed in Section 5.

This skew has two further effects. The first is on measurement. When one label dominates, annotators match often even by chance. A low kappa then becomes hard to read. It may signal genuinely poor annotation. It may also reflect nothing more than the uneven label distribution (Feinstein and Cicchetti, 1990; Byrt et al., 1993). The second effect is on modelling. Labels that are wrong in a systematic way teach the model the same wrong patterns (Frénay and Verleysen, 2014). That model may then be used to label further data. The original blind spot is reproduced at a larger scale, and the resource looks more reliable than it is.

Improving this process requires checks at more than one stage. The priorities below address how labels are produced, checked and retained.

6.2. Report how labels were produced

A dataset can only be reused well if its production is documented. Releases should state who or what assigned the labels, whether human annotators, automatic tools, or a combination, and should include the annotation guidelines. They should also state what judgement the annotators were actually asked to make. A kappa value obtained from a two-class decision does not transfer to a three-class one, yet both appear in a release as a single number.

Releases should describe the annotators as well. Data statements offer an established format, covering the language variety, the speakers, and the annotators' backgrounds (Bender and Friedman, 2018). This allows readers to judge where a dataset applies and where it does not.

Two further items deserve attention. The first is the class distribution, reported together with agreement for each class, so that a single coefficient does not hide weak agreement on the smaller classes. The second is the individual annotations themselves. Retaining each annotator's labels, along with any discarded items, allows later studies to examine where annotators differed rather than relying only on the majority label (Mostafazadeh Davani et al., 2022). The 512 posts removed from MESocSentiment are an example of what is lost when only final labels are kept.

6.3. Evaluate agreement under class imbalance

Future studies should examine how agreement results shift as the proportions of positive, negative and neutral labels change. Byrt et al. (1993) show that a single kappa value is hard to compare across studies, and recommend reporting separate indices for bias and prevalence alongside it. Di Eugenio and Glass (2004) carry this into annotation work, suggesting that kappa be reported together with a prevalence-adjusted figure. Both proposals were developed for two raters and two categories, so their extension to three-class sentiment remains untested. Working this out on

Manglish data would show whether a low kappa signals weak annotation or only the dominance of the neutral class.

6.4. Develop guidance for particles and ambiguity

Guidelines are where the language problems from Section 6.1 can be addressed directly. They should include worked examples showing how the same particle shifts meaning with context, since lah and kan carry attitude that no lexicon entry can capture. Mohammad (2016) offers a useful starting structure, separating the speaker, the target of the opinion, and the attitude expressed, and Kong et al. (2023) show that this structure can be adapted to Malaysian tweets. Guidelines for Manglish could extend it with local particle examples.

Guidelines should also say what to do when an item stays unclear. Forcing a choice between positive, negative and neutral turns genuine ambiguity into an arbitrary label, which then looks like annotator error. Allowing annotators to flag such cases separates disagreement caused by vague instructions, which guidelines can fix, from disagreement caused by the text itself, which they cannot.

6.5. Test whether annotators share errors

Section 5 noted that annotators may agree because they share a mistake, and Section 6.1 described one route by which this happens. Neither agreement coefficients nor majority voting can detect it, so it needs a separate test.

One design would give the same Manglish sample to bilingual annotators, to existing sentiment tools, and to generative models. A second group of bilingual reviewers would then assign reference labels independently and write short explanations for the cases where the first group disagreed or agreed on a doubtful reading. Weber and Plank (2023) show that adding human review to automatic error detection finds more annotation errors than either alone. Weber-Genzel et al. (2024) use explanations and later validity judgements in this way for English textual inference, separating genuine errors from reasonable variation, and the same design could be tested here.

The outcome would show whether majority voting corrects errors or preserves the ones the annotators hold in common. This matters most where humans and automatic tools miss the same Manglish cue, since that is exactly the case a majority vote cannot catch.

7. Conclusion

The four resources reviewed here provide useful Malay-English data, but the evidence supporting their label quality remains incomplete. Some report no agreement coefficient, while others report one for a different task or for only part of the dataset. Examples from these datasets also show that automatic tools can miss positive Malay expressions. This does not establish an overall error rate, but it shows why further validation is needed.

Four steps would strengthen the evidence. Future work should document how labels were produced and retain each annotator's labels, evaluate agreement under

class imbalance, develop guidance for particles and ambiguous cases, and test whether annotators and tools share the same errors. Without these checks, a dataset can look reliable while carrying errors that pass into the models trained on it, and from those models into the data they label next.

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